

Problem Set 7

Instructor: Bill Fefferman

1 Probability theory

Problem 1: A set of k events $A_1, A_2, \dots, A_k$ is called *pairwise independent* if all pairs of events $\{A_i, A_j\}$ are independent, so $\Pr[A_i \cap A_j] = \Pr[A_i] \Pr[A_j]$. More generally we say that these k events are *mutually independent* if for all subsets, $I \subseteq [k]$, of the events, $\Pr[\bigcap_{i \in I} A_i] = \prod_{i \in I} \Pr[A_i]$. Give an example of a collection of events that are *pairwise*, but not *mutually* independent, and explain your answer.

Problem 2: Incredibly, your mother buys 1000 small gifts to give to relatives for the holidays. What's even more amazing is that each of these items is different. There are 100 relatives to distribute the gifts to.

- How many different ways are there to distribute the gifts?
- How many different ways are there to distribute the gifts if she demands that each relative gets an equal number, $1000/100 = 10$ gifts?

Problem 3: Let A and B be two independent events such that $\Pr[A] = 1/4$ and $\Pr[A \cup B] = 2\Pr[B] - \Pr[A]$. Find:

1. $\Pr[B]$
2. $\Pr[A|B]$

Problem 4: Suppose we have a non-negative random variable with expectation value 50. Then:

1. Derive an upper bound for the probability that X is at least 100.
2. Suppose we know that the variance of X is 25. Then can we obtain a better bound for part (1) of this problem?